Answers · Term 4

TERM 4

Teacher copy. Answers were calculated in code and checked. Circumference and Pythagoras answers use the π and √ keys; where π ≈ 3.14 gives a different 1 d.p. answer it is listed as an alternative. Reasoning answers are model responses: accept any correct explanation.

WK 1	Q1	Q2	Q3	Q4	Q5
S1	6 and 7 (because 36 < 37 < 49)	$k = 7$	−2	$x = 53$	60 students
S2	$20	$x = 50$ m	15	$y = 3$ or $y = -3$	$2^3 \times 5$
S3	The base stays 9; only the indices are added: $9^6 \times 9^5 = 9^{11}$, not 81^{11}.	$x = 0.6$ cm	$\frac{m}{4}$	40%	$x = 120$

WK 2	Q1	Q2	Q3	Q4	Q5
S1	Yes: $5^2 + 12^2 = 169 = 13^2$	3^3	The rectangle: rectangle 18 m², square 16 m².	$a = 2.75$ (= $2\frac{3}{4}$)	$x = 60$
S2	$x = 9$	$4(3x - 7)$	69 m²	3^8	$25
S3	$x = 2.5$ cm	270 L	$\frac{3}{10}$	$x = 1$: $y = 8$, $x = 4$: $y = 17$	4^8

WK 3	Q1	Q2	Q3	Q4	Q5
S1	$-20p + 40$	$x = 15$ cm	$2 \times 2 \times 5$	10	$x = 38$
S2	10% profit	$n = 6$	The mean (9) is larger than the median (6) because the outlier 25 pulls the mean up.	9^2	$5n + 33$
S3	$x = 10.8$ cm	$x = 2$ or $x = -2$	$x = 72$	81	360 kg

WK 4	Q1	Q2	Q3	Q4	Q5
S1	$2a^2 + 11a$	40	$5a + 4b$	76 cm²	$n = 9$
S2	0	$3n + 8 = 32$; the number is 8	4^6	$\frac{1}{7}$	$x = 10$ cm
S3	$9(4m - 5)$	No: $8^2 + 9^2 = 145$, but $12^2 = 144$	(2, −2)	$x = 72$ (base angles of an isosceles triangle are equal and the angle sum is 180°)	$99

WK 5	Q1	Q2	Q3	Q4	Q5
S1	The base stays 4; only the indices are added: $4^2 \times 4^4 = 4^6$, not 16^6.	$x = 15$ cm	$k = 1$	$x = 115$	14
S2	$5x$	40%	$2^4 \times 3$	84 cm²	$w = 5$
S3	$3p - 3$	$234	$21	140 m³	$7^1 = 7$

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WK 6	Q1	Q2	Q3	Q4	Q5
S1	$x = 25$ cm	$5x + 32$	\$12	$x = 109$	180 kg
S2	10 and 11 (because $100 < 114 < 121$)	5% profit	7^{16}	Yes: when $x = 2$, $y = 8$.	$x = 6$ or $x = -6$
S3	$9(3y + 7)$	2^5	4	$n = 9$	157.1 cm Also accept 157.0 cm if π ≈ 3.14 is used

WK 7	Q1	Q2	Q3	Q4	Q5
S1	\$240	$a = 2.75$ ($= 2\frac{3}{4}$)	5.66	52 cm	50
S2	$10y + 45$	$h = 3$ (hours)	2^3	$x = 16$ cm	$\frac{13}{20}$ (0.65)
S3	$a^2 + 9a$	$2n + 13 = 21$; the number is 4	Sam added first; × must be done before +. Correct answer 30.	$x = 131$	6^8

WK 8	Q1	Q2	Q3	Q4	Q5
S1	The base stays 3; only the indices are added: $3^4 \times 3^4 = 3^8$, not 9^8.	$5x + 12$	$\frac{7}{10}$	25 cm^2	\$48
S2	\$8	$k = 9$	4	$7(a + 2)$	$x = 13$ cm
S3	10%	$2 \times 3^2 \times 5$	10	No: $4^2 + 5^2 = 41$, but $6^2 = 36$	$5m$

WK 9	Q1	Q2	Q3	Q4	Q5
S1	$x = 3.6$ m	6^3	$x = 7$ or $x = -7$	25	$x = 62$
S2	$40m + 32$	75 kg	The rectangle: rectangle 20 m^2, square 16 m^2.	$x = 8$ m	$t = 6$
S3	$k = 6$	$x = 6.5$ m	26	50% profit	15%

WK 10	Q1	Q2	Q3	Q4	Q5
S1	\$63	$a = 6.5$ ($= 6\frac{1}{2}$)	$x = 52$	−4	30 000 m^2
S2	$2n + 4 = 26$; the number is 11	2^6	2600	$6(x + 4)$	9 and 10 (because $81 < 87 < 100$)
S3	$x = 5$ cm	\$160	No: P(blue) = $\frac{1}{3}$, which is less than $\frac{1}{2}$.	$x = 97$	$6p^2$